October 12, 2026

Six burdens with no tagged result in a library of 25 Lyme disease and Long COVID simulators

Policy brief. Uri Kartoun, PhD, DBbun LLC. Counts rechecked October 11, 2026. Independent entry in the TOPx HHS Tech Sprint; not endorsed by HHS or NIH. Built from the 25 simulators in the DBbun library (12 Lyme disease, 13 Long COVID), derived from 23 papers and two reports.

Finding

Across the 25 simulators, 67 numeric results were tagged with one of 16 burden categories. 49 of those results (73%) fall in just three categories: work and income, direct costs of care, undercounted by official surveillance. Six categories have no tagged result in any simulator, for either condition: care continuity risk, caregiving and household labor, cognitive impact, congenital/parent-to-child transmission, longer-term opportunity costs, psychological burden. That shows where this library's numbers are thin. It does not show that these burdens are small, or that they were never studied elsewhere.

Burden category	Lyme	Long COVID	Total
Work and income	2	15	17
Direct costs of care	13	3	16
Undercounted by official surveillance	3	13	16
Diagnostic delay	6	none	6
Barriers to reaching care	4	none	4
Overall quality of life	none	3	3
Administrative burden	2	none	2
Harm from misdiagnosis-driven treatment	1	none	1
Medical disbelief and dismissal	1	none	1
Uncoordinated, fragmented care	none	1	1
Care continuity risk	**none**	**none**	**none**
Caregiving and household labor	**none**	**none**	**none**
Cognitive impact	**none**	**none**	**none**
Congenital/parent-to-child transmission	**none**	**none**	**none**
Longer-term opportunity costs	**none**	**none**	**none**
Psychological burden	**none**	**none**	**none**

Numbers are results per burden category (each result has at most one tag). Bold rows have none. Counts are tagged numeric results, not numbers of simulators.

Why it matters

- **Claims and codes miss it.** In 14 of 25 simulators the saved record flags a source warning that claims data or ICD-10 codes undercount the condition; in 20 of 25 it flags a gap in the care or evidence system.
- **What is counted is lopsided.** In this library, work and income, direct costs of care, and undercounting account for 49 of 67 tagged numeric results (73%). The six categories listed above have no tagged numeric result in these 25 simulators.

What measuring the six would look like

- **Care continuity risk:** Share of care that depends on one person.
- **Caregiving and household labor:** Caregiver hours per week.
- **Cognitive impact:** Task completion time; share needing help.
- **Congenital/parent-to-child transmission:** Transmission rate; number of children affected.
- **Longer-term opportunity costs:** Years of schooling disrupted; tutoring hours; projected career earnings lost.
- **Psychological burden:** Anxiety or depression prevalence.

Suggested next steps (the author's suggestions, not findings)

- Ask whether an existing federal survey could add an item on caregiver hours per week and one on years of schooling disrupted. The other four would need a review of clinical relevance and feasibility first.

How far to trust this

Dr. Kartoun chose the papers, not through a systematic review, so the counts describe this library, not all Lyme disease or Long COVID research. An AI model (Claude Sonnet 5, claude-sonnet-5) assigned each tag; code checked them against a fixed list. "None" means no numeric result in the saved simulator outputs was assigned that category. It does not mean the burden is absent or unstudied elsewhere. This brief summarizes published sources and saved simulator outputs; it does not analyze individual patient records. I directed this brief's preparation and reviewed and modified AI-generated text produced with Claude Code (Claude Sonnet 5.5, claude-sonnet-5-5). Live counts: public demo, Simulation library.